PMx Weekly Digest: April 04, 2026

Paper 1: The Potential of Digital Twins for Pediatric Rare Diseases

Bibliographic Information

- **Title**: The Potential of Digital Twins for Pediatric Rare Diseases
- **Authors**: Malik-Sheriff RS, et al.
- **Journal**: CPT: Pharmacometrics & Systems Pharmacology, 2026
- **DOI**: https://doi.org/10.1002/psp4.70234

Quick Take

Pharmacometricians, pediatric drug developers, and rare disease researchers should read this comprehensive review outlining how hybrid mechanistic-AI Digital Twins can overcome ethical and logistical barriers in pediatric rare disease research through in silico trials, synthetic control arms, and precision dosing simulations. The paper synthesizes regulatory guidance (ICH M15, FDA NAMs), technical methodologies (federated learning, synthetic data generation), and pediatric-specific ethical considerations. Essential reading for those navigating the transition from population-based MIDD to individualized virtual patient models.

Executive Summary

This position paper articulates a strategic framework for deploying Digital Twins (DTs) in pediatric rare diseases, where traditional clinical trial methodologies fail due to ethical constraints, small patient numbers, and developmental physiological complexity. The authors propose hybrid mechanistic-AI/ML architectures that combine interpretable quantitative systems pharmacology (QSP) cores with flexible machine learning components to address data sparsity while maintaining biological plausibility. The work synthesizes regulatory momentum (FDA/EMA guidance, ICH M15 framework), technical methodologies (synthetic data generation, federated learning, uncertainty quantification), and ethical-legal considerations specific to pediatric populations, positioning DTs as transformative tools for precision diagnostics, personalized therapy optimization, and in silico clinical trials in historically underserved therapeutic areas.

⚡ Methodological Snapshot

The paper advocates for hybrid Digital Twin architectures that integrate mechanistic systems pharmacology models with machine learning components to address the dual challenges of biological interpretability and data scarcity in pediatric rare diseases. The mechanistic core—typically formulated as systems of ordinary differential equations representing biochemical, pharmacokinetic, or physiological pathways—provides a structured, causal framework grounded in prior biological knowledge. This is coupled with ML components (deep ensembles, Bayesian neural networks, Gaussian processes, or neural ODEs) that capture residual errors, unknown dynamics, or high-dimensional patterns not explicitly represented in the mechanistic structure. This hybrid approach enables the model to learn from sparse, heterogeneous datasets while maintaining the biological plausibility and regulatory acceptability necessary for clinical decision-making.

The methodology emphasizes rigorous uncertainty quantification through Bayesian hierarchical modeling and ensemble methods, identifiability analysis using profile likelihoods and Sobol sensitivity indices, and validation strategies tailored to synthetic data generation—including statistical fidelity metrics, differential privacy for disclosure risk, and biological coherence assessment across multi-omics modalities.

⊠⊠ Structural Model Breakdown

The Digital Twin architecture comprises three primary structural configurations: (1) Pure mechanistic models utilizing systems of differential equations to represent biochemical, pharmacokinetic ($PBPK$), or physiological (QSP) pathways; (2) Pure AI/ML models using deep neural networks or ensemble methods trained on large-scale patient data for pattern recognition and prediction; and (3) Hybrid mechanistic-ML models—the recommended approach—where a mechanistic core provides biophysical constraints and interpretability while ML components (multilayer perceptrons, Gaussian processes, Bayesian neural networks, or neural ODEs) model residuals, unknown dynamics, or high-dimensional multi-omics relationships. In the hybrid configuration, the mechanistic backbone typically consists of compartmental ODEs representing drug absorption, distribution, metabolism, and elimination, or systems biology models of disease pathways (e.g., glycogen accumulation in Pompe disease). ML layers are constrained to capture deviations from mechanistic predictions or to infer latent variables from high-dimensional inputs (genomic, imaging, clinical data). The coupling between components requires careful architectural decisions: ML can model additive residuals, multiplicative corrections, or entirely unmodeled pathways, with orthogonality constraints recommended to prevent parameter absorption. Developmental stage-specific parameters must be incorporated to account for ontogeny-driven changes in clearance (CL), volume of distribution (V_d), and receptor expression across pediatric age groups.

Detailed Methodological Analysis

Modeling Approach

Hybrid mechanistic-AI/ML framework combining quantitative systems pharmacology (QSP), physiologically-based pharmacokinetic ($PBPK$), or quantitative systems toxicology (QST) mechanistic cores with adaptive ML components (deep neural networks, Gaussian processes). Mechanistic models provide structure, interpretability, and biophysical constraints; ML components handle unknown dynamics, residuals, and high-dimensional pattern recognition. Alternative approaches include pure mechanistic models (systems of differential equations) and pure AI/ML models (data-driven predictions), though these are less suitable for rare disease contexts. Software standards include SBML for mechanistic models and ONNX for ML model interoperability.

Data Sources

Multi-omics datasets (genomics, transcriptomics, epigenomics, proteomics) integrated with longitudinal clinical phenotypes and real-world evidence from rare disease registries (Orphanet, RDCRN, European Reference Networks). Synthetic data generated via transfer learning from

common diseases with overlapping biological mechanisms. Federated learning approaches to leverage disparate institutional datasets without centralized aggregation. Pediatric-specific challenges include limited neonatal/infant data, geographic dispersion of small cohorts, and incompatible consent frameworks.

Estimation Methods

Bayesian calibration with posterior predictive checks for mechanistic model components. Deep ensembles of multilayer perceptrons, Bayesian neural networks, Gaussian process surrogates, and neural differential equations for ML components. Profile likelihood methods for structural and practical identifiability analysis. Sobol global sensitivity analysis to quantify parameter contributions to output variability. Feature attribution methods (SHAP) for post-hoc interpretability of ML predictions.

Model Evaluation

Validation of synthetic data through distributional similarity metrics (Hellinger distance), pairwise correlation differences, and task-specific performance on held-out real data. Cross-modal biological coherence assessment (e.g., pathway enrichment) for multi-omics synthetic datasets. Credibility assessment following ICH M15 framework for regulatory acceptance. Disclosure risk evaluation using differential privacy methods. Counterfactual testing to probe prediction changes under hypothetical interventions.

Covariate Analysis

Developmental-stage-specific parameters accounting for ontogeny-driven changes in pharmacokinetics, pharmacodynamics, and immune function. Allometric scaling considerations for pediatric populations spanning neonates to adolescents. Integration of multi-omics covariates (genomic variants, transcriptomic signatures, proteomic profiles) with clinical phenotypes for patient stratification and personalized predictions.

Scientific Context & Motivation

Rare diseases affect over 300 million people globally, with 75% manifesting in childhood and only 5% having approved treatments. Pediatric rare disease research faces unique methodological barriers: extreme data scarcity due to small patient numbers and geographic dispersion; ethical constraints limiting empirical experimentation in children; developmental physiological diversity across neonatal, infant, child, and adolescent stages; and diagnostic odysseys with <35% diagnostic rates despite whole-genome sequencing. Traditional clinical trial methodologies fail in this context due to recruitment impossibility and the prohibitive expense of n=1 approaches. Current adult reference models are inadequate as infant, child, and adolescent physiology fundamentally differs from adults in drug metabolism, immune function, and organ development. Digital Twins address this knowledge gap by enabling in silico hypothesis testing, precision diagnostics, and personalized therapy optimization where empirical studies are infeasible, representing a paradigm shift from population-averaged to individualized model-informed drug development.

Statistical Rigor Assessment

The paper appropriately emphasizes uncertainty quantification (UQ) as critical for hybrid model credibility, acknowledging multiple uncertainty sources: aleatory (observation noise), epistemic (limited data, incomplete knowledge), and structural (model simplification errors). Bayesian hierarchical models and ensemble methods are recommended for generating parameter and output distributions, with calibration ensuring predicted uncertainties reflect real-world performance. Identifiability analysis—both structural (theoretical) and practical (empirical via profile likelihoods)—is highlighted as essential to ensure parameters are learnable from sparse rare disease data. However, the paper notes inherent tensions in hybrid models where ML components may absorb mechanistic dynamics or replicate known behaviors, undermining unique parameter inference. Sample size considerations are addressed through synthetic data generation and federated learning, though validation of synthetic data remains challenging. Missing data handling is implicitly addressed through multi-omics integration and generative AI augmentation, though specific statistical methods for missingness mechanisms are not detailed. Sensitivity analysis via Sobol indices provides robustness assessment for mechanistic components.

⊠ Key Findings

Hybrid mechanistic-AI/ML architectures offer an optimal balance of interpretability and adaptability for rare disease modeling, where pure mechanistic models lack flexibility and pure ML models lack biological grounding. Synthetic data generation through transfer learning from common diseases can augment sparse pediatric rare disease datasets, though rigorous validation using distributional similarity metrics (Hellinger distance), disclosure risk assessment (differential privacy), and task-specific performance testing is mandatory to prevent hallucination and bias amplification. Regulatory momentum is substantial: FDA and EMA guidance supports in silico trials and NAMs (New Approach Methodologies), with ICH M15 providing harmonized credibility assessment frameworks for complex models. Pediatric DTs require developmental-stage-specific parameters reflecting ontogeny of drug metabolism, immune function, and physiology—factors absent in adult reference models. Federated learning emerges as a critical infrastructure solution to overcome data fragmentation caused by geographic dispersion of rare disease cohorts and incompatible consent frameworks across institutions. Uncertainty quantification must address multiple sources: observation noise, limited data, structural model error, and epistemic uncertainty from unknown dynamics, requiring Bayesian hierarchical approaches and ensemble methods applied to the full hybrid system.

💡 Clinical & Regulatory Implications

Digital Twins enable precision dosing simulations that account for developmental pharmacokinetics and ontogeny-driven changes in drug metabolism, critical for pediatric populations where empirical dose-finding trials are ethically constrained. They facilitate in silico trials with synthetic control arms, potentially accelerating rare disease drug development where recruitment is prohibitive. For diagnosis, DTs integrating multi-omics (genomics, transcriptomics, proteomics) with clinical phenotypes can accelerate recognition of ultra-rare patterns and predict therapeutic

responses to immunosuppressants and enzyme replacement therapies (e.g., avalglucosidase alfa in Pompe disease). Regulatory agencies (FDA, EMA) increasingly accept DT-derived evidence under MIDD frameworks, with the ICH M15 guideline providing harmonized credibility assessment criteria. However, implementation requires careful consideration of pediatric consent dynamics, GDPR compliance for genetic data, and the transition of data stewardship from parents to patients as children mature.

Strengths & Limitations

Strengths

- Comprehensive integration of technical, regulatory, and ethical dimensions of pediatric rare disease modeling
- Balanced assessment of pure mechanistic vs. pure AI vs. hybrid approaches with clear rationale for the latter
- Recognition of pediatric-specific physiological and ethical complexities often overlooked in general DT literature
- Practical framework for uncertainty quantification combining Bayesian methods with global sensitivity analysis
- Strategic emphasis on federated learning to address data fragmentation without compromising privacy
- Alignment with emerging regulatory standards (ICH M15, FDA NAMs) providing actionable guidance for implementation
- Inclusion of specific validation metrics for synthetic data (Hellinger distance, differential privacy) beyond generic statements
- Acknowledgment of identifiability challenges in hybrid models—a critical but often neglected methodological issue

Limitations (Acknowledged by Authors)

- Data scarcity and poor standardization of multi-omics and longitudinal clinical data in pediatric rare diseases
- Potential for hallucination and bias amplification in synthetic data generation
- Computational and resource intensity of constructing and validating hybrid DTs
- Regulatory uncertainty regarding acceptance criteria for hybrid mechanistic-ML models
- Ethical and legal constraints (GDPR, pediatric consent) limiting longitudinal data collection and multi-generational genetic analysis
- Weak identifiability of ML components and risk of parameter absorption between mechanistic and ML layers
- Limited availability of coupled multi-omics datasets for rare diseases
- Challenges in validating synthetic data for biological coherence beyond statistical similarity

Limitations (Expert Review)

- Lack of specific quantitative case studies demonstrating full hybrid model implementation with parameter estimates and predictive performance metrics

- Absence of detailed mathematical specifications for the proposed hybrid architecture (e.g., how mechanistic and ML components are mathematically coupled)
- Insufficient discussion of model selection criteria when choosing between pure mechanistic, pure ML, and hybrid approaches for specific rare diseases
- Limited consideration of model updating procedures as new longitudinal data arrives (adaptive/online learning)
- Underemphasis on the validation burden for regulatory acceptance—specific criteria for establishing credibility of synthetic control arms remain undefined
- Geographic bias toward US/EU regulatory frameworks with limited discussion of other major markets (Japan, China)
- Potential overestimation of current AI/ML capabilities for handling extreme data scarcity ($n < 10$) common in ultra-rare diseases

Generalizability

The framework is broadly applicable across pediatric rare diseases, though specific implementations require disease-specific mechanistic knowledge and pediatric pharmacokinetic data. Generalizability is limited by the current scarcity of validated pediatric ontogeny models and the need for disease-specific synthetic data validation. The regulatory recommendations apply primarily to FDA/EMA jurisdictions, though ICH M15 provides international harmonization.

Figures & Tables

- **Figure 1**: Conceptual diagram illustrating the multifaceted challenges (data scarcity, ethical constraints, developmental complexity) and enabling technologies (synthetic data generation, federated learning, hybrid modeling) for constructing pediatric rare disease Digital Twins.
  - *Significance*: Provides a high-level strategic overview of the DT development pipeline, emphasizing the central role of generative AI for data augmentation and the iterative refinement of virtual patient models with longitudinal clinical data. Essential for understanding the paper's proposed workflow and the intersection of technical, regulatory, and ethical domains.

Future Directions

Immediate research priorities include developing automated methodologies for generating mechanistic models directly from multi-omics data to reduce the resource burden of DT construction, and establishing standardized validation frameworks for synthetic pediatric datasets that assess both statistical fidelity and biological coherence across modalities. Federated learning networks (such as the International Precision Child Health Partnership) must mature to enable cross-institutional model training without centralized data sharing, addressing GDPR and pediatric consent constraints. Methodological work is needed to resolve identifiability conflicts between mechanistic and ML components, potentially through constrained optimization or orthogonal parameterization strategies. Longitudinal studies tracking DT predictive accuracy across developmental stages—from neonates through adolescence—are essential to validate ontogeny-specific parameterizations. Regulatory science should focus on defining acceptance criteria for hybrid

model credibility assessment under ICH M15, particularly regarding uncertainty calibration and synthetic control arm validation. Finally, ethical frameworks must evolve to accommodate dynamic consent mechanisms and multi-generational genetic data stewardship in rare disease contexts.

Expert Commentary

This paper arrives at a critical inflection point in pharmacometrics, as the field transitions from population-based nonlinear mixed-effects models toward individualized virtual patient representations. The authors astutely recognize that pure AI/ML approaches lack the mechanistic grounding necessary for regulatory acceptance in drug development, while purely mechanistic models cannot adapt to the heterogeneity and data scarcity of rare diseases. The hybrid approach they advocate—anchoring ML components to validated QSP structures—mirrors emerging best practices in Bayesian neural ODEs and physics-informed machine learning. However, the identifiability challenges they highlight between mechanistic and ML components represent a fundamental unsolved problem; when neural networks model residuals or absorb dynamics that should constrain mechanistic parameters, we risk losing the very interpretability that makes these models regulatory-friendly. The emphasis on federated learning for pediatric data is particularly salient, as geographic dispersion and consent complexity have historically fragmented rare disease datasets. From a regulatory perspective, the reference to ICH M15 (finalized 2026) and FDA's New Approach Methodologies ($NAMs$) signals that DTs are transitioning from academic curiosity to regulatory science. Yet I caution that the validation burden for hybrid models remains undefined—how do we validate a model that changes its structure as it learns? The pediatric-specific ethical considerations (dynamic consent, parental vs. patient stewardship) are often overlooked in technical pharmacometrics literature and represent essential reading for anyone modeling pediatric populations. The synthetic data discussion is pragmatic but understates the risk of hallucinated biological pathways; validation must extend beyond statistical fidelity to mechanistic plausibility. Overall, this paper serves as both a roadmap and a warning: DTs offer unprecedented opportunity for pediatric rare diseases, but require rigorous uncertainty quantification, transparent validation, and careful ethical stewardship to realize their potential without compromising patient safety or scientific rigor.

Bottom Line

Digital Twins represent an evolutionary leap from traditional population-based MIDD toward individualized, dynamic virtual patient representations that integrate mechanistic systems pharmacology with adaptive AI/ML components. For pediatric rare diseases—where ethical constraints, developmental physiology, and data scarcity converge—hybrid modeling approaches offer a viable pathway to precision dosing and in silico trial design, provided rigorous uncertainty quantification, federated data infrastructure, and pediatric-specific ethical frameworks are established. Practicing pharmacometricians should prepare for regulatory acceptance of synthetic control arms and hybrid QSP-ML models under the ICH M15 framework, while remaining

vigilant about identifiability challenges and validation requirements when mechanistic and black-box components are coupled.

⊠ Figures

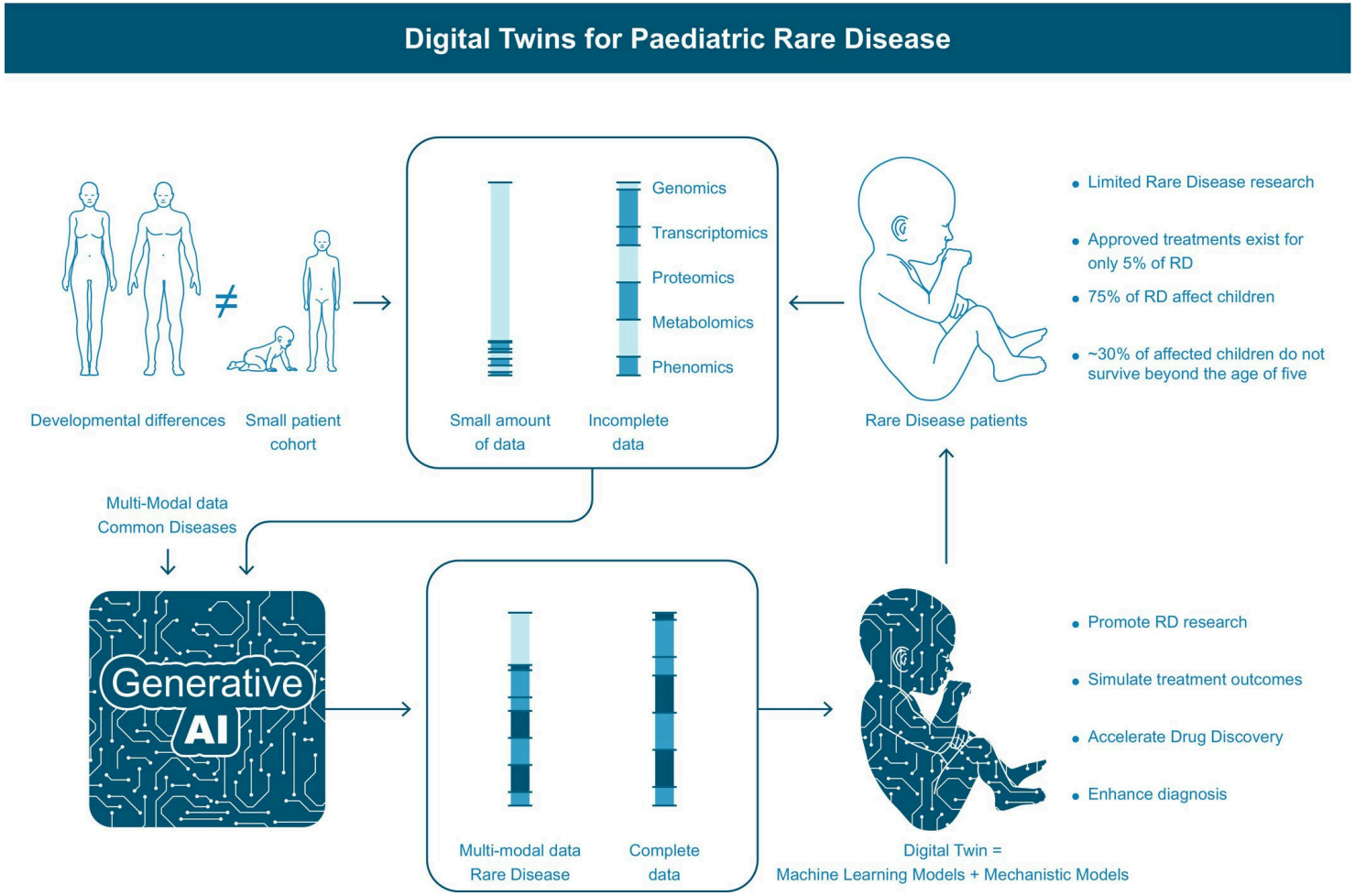

Figure 1: FIGURE 1: Challenges and opportunities in building rare disease digital twins.